

1. Administrative & Study Identification

Full Study Title: A Randomized, Open-label, Phase 3 Study of MK-2870 vs. Platinum Doublets in Participants With EGFR-mutated, Advanced Nonsquamous Non-small Cell Lung Cancer (NSCLC) Who Have Progressed on Prior EGFR Tyrosine Kinase Inhibitors **Short Title/ Acronym:** Sacituzumab Tirumotecan (MK-2870) Versus Pemetrexed and Carboplatin Combination Therapy in Participants With Epidermal Growth Factor (EGFR)-Mutated, Advanced Nonsquamous Non-small Cell Lung Cancer (NSCLC) Who Have Progressed on Prior EGFR Tyrosine Kinase Inhibitors (MK-2870-009) **Protocol Number:** MK-2870-009 **NCT Number:** NCT06305754 **Posting ID:** 774835801660864471 **Sponsor Name:** Merck **Company:** Merck **Investigational Product:** Sacituzumab tirumotecan (MK-2870) **Phase of Development:** PHASE3 **Trial Status:** RECRUITING **Disease Areas:** Non-small Cell Lung Cancer (NSCLC) **Medical Monitor:** [Name and Contact Information]

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate whether sacituzumab tirumotecan is better than platinum-based doublet chemotherapy (pemetrexed and carboplatin) in regard to progression-free survival (PFS) and overall survival (OS). **Secondary Objectives:** To evaluate the objective response rate (ORR), disease control rate (DCR), time to response (TTR), duration of response (DOR), change from baseline in EORTC QLQ-C30, and to assess the safety and tolerability of the treatments.

2.2 Background & Rationale To address the unmet medical need in patients with epidermal growth factor receptor (EGFR)-mutated advanced non-squamous non-small cell lung cancer (NSCLC) whose disease has continued to progress following prior treatment with EGFR tyrosine kinase inhibitors (TKIs). Sacituzumab tirumotecan is an investigational trophoblast cell-surface antigen 2 (TROP2)-directed antibody-drug conjugate (ADC) designed to offer improved clinical outcomes over standard-of-care chemotherapy in this refractory setting.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Active Comparator (Pemetrexed plus Carboplatin) **Blinding:** Open-label (Single, Unblinded) **Randomization:** 1:1

3.2 Planned Enrollment & Duration Targeted Enrollment: 520 participants **Number of Sites:** Multinational / Global **Estimated**

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Histologically or cytologically confirmed diagnosis of advanced-stage nonsquamous non-small cell lung cancer (NSCLC). 2. Participants who have adverse events (AEs) due to previous anticancer therapies must have recovered to Grade ≤ 1 or baseline. 3. Participants who are Hepatitis B surface antigen (HBsAg) positive are eligible if they have received Hepatitis B virus (HBV) antiviral therapy for at least 4 weeks and have undetectable HBV viral load. 4. Participants with history of Hepatitis C virus (HCV) infection are eligible if HCV viral load is undetectable. 5. Human immunodeficiency virus (HIV)-infected participants must have well controlled HIV on antiretroviral therapy. 6. Life expectancy of at least 3 months.

4.2 Exclusion Criteria 1. Predominantly squamous cell histology NSCLC. 2. History of second malignancy, unless potentially curative treatment has been completed with no evidence of malignancy for 3 years. 3. Grade ≥ 2 peripheral neuropathy. 4. History of documented severe dry eye syndrome, severe Meibomian gland disease and/or blepharitis, or severe corneal disease that prevents/delays corneal healing. 5. Active inflammatory bowel disease requiring immunosuppressive medication or previous history of inflammatory bowel disease. 6. Uncontrolled, or significant cardiovascular disease or cerebrovascular disease. 7. Received prior radiotherapy within 2 weeks of start of study intervention, or radiation-related toxicities, requiring corticosteroids. 8. Received a live or live-attenuated vaccine within 30 days before the first dose of study intervention. Administration of killed vaccines is allowed. 9. Received radiation therapy to the lung that is >30 Gray within 6 months of the first dose of study intervention. 10. Known active central nervous system metastases and/or carcinomatous meningitis. 11. Active infection requiring systemic therapy. 12. History of (noninfectious) pneumonitis/interstitial lung disease that required steroids or has current pneumonitis/interstitial lung disease. 13. HIV-infected participants with a history of Kaposi's sarcoma and/or Multicentric Castleman's Disease. 14. Concurrent active HBV and HCV infection. 15. History of allogeneic tissue/solid organ transplant. 16. Participants who have not adequately recovered from major surgery or have ongoing surgical complications.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Sacituzumab tirumotecan (MK-2870) OR Pemetrexed plus Carboplatin (administered via standard oncology cycles). **Route of**

Administration: Intravenous (IV) **Duration of Treatment:** Treatment continues until any of the criteria for discontinuation of study intervention are met (e.g., disease progression or unacceptable toxicity). Patients will be followed for overall survival for up to approximately 51 months.

5.2 Concomitant & Prohibited Therapy Permitted: Standard concomitant supportive care medications (e.g., H1/H2 receptor antagonists, paracetamol, dexamethasone, steroid mouthwash). **Prohibited:** Other investigational agents or concurrent systemic anti-cancer therapies outside the study protocol.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Medical History, Tumor Mutational Profiling, Baseline Scans, Labs
Baseline	Day 0	Randomization, First Dose
Treatment Cycles	Per Protocol	Dosing, Safety Labs, AE Monitoring, Objective Response Assessment via Imaging
End of Treatment	N/A	Treatment Discontinuation, Final Vital Signs, Exit Interview
Follow-up	Up to 51 mo.	Post-treatment survival follow-up, subsequent anti-cancer therapy tracking

7. Outcome Measures (Endpoints)

7.1 Primary Endpoints Definition: 1. Progression-free survival (PFS): Time from randomization to the first documented disease progression per Response Evaluation Criteria in Solid Tumors Version 1.1 (RECIST 1.1) based on blinded independent central review (BICR) or death due to any cause, whichever occurs first. 2. Overall survival (OS): Time from randomization to death due to any cause. **Measurement Method:** Assessed by BICR and Investigator per RECIST v1.1 for PFS.

7.2 Secondary & Exploratory Endpoints 1. Objective Response Rate (ORR). 2. Disease Control Rate (DCR). 3. Duration of Response (DOR) and Time to Response (TTR). 4. Change from Baseline in European Organization for Research and Treatment of Cancer Quality of Life Questionnaire-Core 30 (EORTC QLQ-C30) Global Health Status (Item 29) and Quality of Life (Item 30) Combined Score. 5. Incidence and severity of Adverse Events (AEs).

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous monitoring from informed consent through the end of the safety follow-up period. **Serious Adverse Events (SAEs):** Required to be reported to the sponsor and relevant regulatory authorities within 24 hours of site awareness. **Data Monitoring Committee (DMC):** Independent Data

Monitoring Committee (eDMC) to assess unblinded safety and efficacy data during the trial.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) for primary efficacy endpoints (PFS and OS); Safety Set for all participants receiving at least one dose of study medication. **Sample Size Justification:** Target enrollment of \$N=520\$ participants, powered to detect a statistically significant difference in PFS and OS between the investigational and active comparator arms. **Handling of Missing Data:** Standard oncology guidelines incorporating right-censoring for PFS and OS endpoints for patients lost to follow-up or without an event.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Regular clinical site monitoring visits (onsite and remote) to ensure protocol adherence and patient safety. **Audit Trail:** Continuous data validation, eCRF locking, and resolution of data clarification forms prior to database lock.

11. Study Identifiers & Governance

Primary ID: {{MK-2870-009}} Posting ID: {{774835801660864471}}
Registry Number: {{NCT06305754}} Sponsor/Lead Organization: {{Merck}} Company: {{Merck}} Trial Status: {{RECRUITING}} Phase: {{PHASE3}} Disease Areas: {{Non-small Cell Lung Cancer (NSCLC)}} Study Title: {{Sacituzumab Tirumotecan (MK-2870) Versus Pemetrexed and Carboplatin Combination Therapy in Participants With Epidermal Growth Factor (EGFR)-Mutated, Advanced Nonsquamous Non-small Cell Lung Cancer (NSCLC) Who Have Progressed on Prior EGFR Tyrosine Kinase Inhibitors (MK-2870-009)}} Official Regulatory Title: {{A Randomized, Open-label, Phase 3 Study of MK-2870 vs. Platinum Doublets in Participants With EGFR-mutated, Advanced Nonsquamous Non-small Cell Lung Cancer (NSCLC) Who Have Progressed on Prior EGFR Tyrosine Kinase Inhibitors}}

12. Clinical Framework & Semantic Concepts

Primary Condition: {{EGFR-mutated Advanced Nonsquamous Non-small Cell Lung Cancer (NSCLC)}} Preferred UMLS Name: {{EGFR Gene Positive Non-Small Cell Lung Carcinoma}} MeSH Code: {{D002275}} SNOMED ID: {{254635002}} Mapped Drug Name: {{paracetamol}} ATC Code: {{N02BE01}} Diagnosis Threshold: {{Disease progression on prior EGFR Tyrosine Kinase Inhibitors (TKIs)}} Medical

Methodology: {{Antibody-Drug Conjugate vs. Platinum-Based Doublet Chemotherapy}} Optimization Target: {{Progression-Free Survival (PFS) and Overall Survival (OS)}}

13. Study Procedures & Methodology

Management Pathway: {{Standard Oncology Management for Refractory NSCLC}} Education Protocol (HCP): {{Protocol-specific training on Investigational Product preparation, administration, and safety management}} Education Protocol (Patient): {{Informed consent process, treatment schedule overview, and AE self-reporting instructions}} Quality Audit Mechanism: {{Centralized imaging review by BICR and routine clinical data monitoring}}

14. Observation & Follow-Up Schedule

Total Duration: {{Up to approximately 51 months for Overall Survival tracking}} Onsite Visit Cadence: {{According to treatment cycle schedule (e.g., Q3W)}} Remote Monitoring Frequency: {{During survival follow-up phase after treatment discontinuation}} Checklist Review Interval: {{Per cycle clinical assessments and imaging every 6-9 weeks}}

15. Sign-off & Approval

Role	Name	Signature	Date	:---	:---	:---	:---
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				